

2024



World Kinect Boosts Application Performance, Drives Infrastructure Resiliency, and Modernizes Applications by Collaborating with AWS

Learn how World Kinect migrated 22 data centers and its Oracle E-Business Suite to AWS and upskilled its employees on AWS, preventing revenue loss by reducing downtime, and boosting productivity and resiliency.

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20%

improvement in Oracle application performance

400%

increase in software deployment



2

50%

Mean Time to Restore

Overview

[World Kinect Corporation](#) is a global energy management company, offering a broad suite of solutions across the fuel and energy product spectrum. To reduce costs and increase efficiency, the company migrated its 22 data centers and critical applications to AWS, including its Oracle E-Business Suite ERP.

By working together to provide technical and advisory consultation, timely support, and training services, AWS experts and instructors helped World Kinect complete migrations to AWS. World Kinect reduced downtime, increased Oracle application performance by 20 percent, and boosted software development by 400 percent.

Opportunity | Migrating Data and Business-Critical Applications to AWS

World Kinect Corporation is a global energy management company that offers a broad suite of solutions across the fuel and energy product spectrum. The company generated nearly \$60 billion in annual revenue in 2022 and serves more than 150,000 customers worldwide in the aviation, marine, and land transportation sectors. It also supplies natural gas and power in the United States and Europe along with a growing suite of other sustainability-related products and services.

In 2019, World Kinect began a cloud transformation journey, modernizing its IT footprint in response to growing business requirements. Josh McLean, chief information officer at World Kinect, explains, "Over 10 years, we acquired



a significant number of companies around the world, and we needed to integrate new people, processes, and technology into World Kinect. We saw the cloud as an opportunity to scale our IT environment to meet business growth.” The company’s modernization strategy also sought to address the limitations of its on-premises data centers. “We had an aging architecture and application space, and an older network with performance issues that caused outages and business disruption,” says McLean.

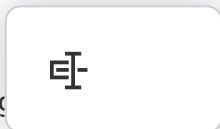
To solve these problems, World Kinect developed a five-year plan to adopt a cloud-first approach, involving the shutdown of all 22 of its physical data centers and the migration of business applications to the cloud. World Kinect chose to work closely with Amazon Web Services (AWS) to support its transformation journey. “We wanted to become a digital and cloud-first organization, and that meant moving all our applications to AWS to automate manual processes, gain centralized access to data, and prepare for future modernization,” McLean says.

After migrating half of its application environment to AWS, World Kinect realized tangible operational cost savings. However, the company still needed to migrate one of its most business-critical systems: the Oracle E-Business Suite, which runs World Kinect’s core enterprise resource planning (ERP) functions. World Kinect sought to move its Oracle solution to AWS for easier scalability and increased business resiliency. Michael Villis, vice president of domain architecture – infrastructure at World Kinect, says, “We waited to migrate our Oracle E-Business Suite because of its size and complexity, with more than 30 custom applications integrated with it. These critical systems include financial operations such as customer invoicing and billing, pricing, and customer self-service.”

World Kinect’s leaders knew this migration needed to be smooth, fast, and efficient—any downtime could risk losing millions of dollars in revenue. The company wanted to equip its employees with the necessary knowledge for both the migration and post-migration operation in the cloud. Specifically, the company knew its transformation would require organizational changes and new roles. “To become a cloud-first organization, you’re not done after the migration,” says Villis. “It’s an ongoing exercise and investing in your own talent is a key part of that.”

Solution | Streamlining Migration and Upskilling Employees with AWS

To ensure a smooth migration and meet its employee training goals, World Kinect initially engaged [Enterprise Support](#). “Our objective was to bring in AWS expertise to automate platform operations such as



environment cloning and failover, and provide us with confidence on the platform's reliability," says Villis. Following this engagement, World Kinect participated in the [AWS Migration Acceleration Program \(MAP\)](#) led by [AWS Professional Services](#). Throughout the process, multiple AWS teams collaborated seamlessly, with activities orchestrated by an AWS account team. "We prioritized performance and reliability, while ensuring appropriate architecture design. AWS not only gave us confidence in these areas, but also offered specialized support for Oracle systems," adds Villis.

Over an 18-month period, AWS helped World Kinect migrate its Oracle E-Business Suite to AWS. As a result, World Kinect experienced increased resilience and performance through services such as [Application Load Balancer](#) for load-balancing application traffic and [Amazon Elastic File System \(Amazon EFS\)](#) to automatically provision and manage application files. Additionally, the company uses multiple [Availability Zones \(AZs\)](#) to provide high availability for application and database tiers.

During and after the migration, World Kinect collaborated with [AWS Training and Certification](#) to implement training initiatives for 85 percent of its IT workforce. The AWS Learning Needs Analysis played an important role in identifying cloud skill gaps, leading to comprehensive hands-on AWS training. This included engaging activities such as AWS Jam sessions, Immersion Days, and GameDays, which helped World Kinect employees build confidence in what they've learned and re-establish learning through real-world use cases. "The Immersion Days and Jam sessions were key in helping employees not only build capabilities directly related to their jobs, but also learn more about AWS services they were unfamiliar with," says Villis. AWS instructors also helped World Kinect build the right skills to facilitate organizational change. In addition, World Kinect engineers can rely on AWS Enterprise Support to quickly contact subject matter experts via phone or chat for guidance.

Outcome | Protecting Revenue, Boosting Application Performance, and Building Internal AWS Skills

Through joint efforts with AWS, World Kinect minimized Oracle E-Business Suite downtime during the migration. With its [Oracle E-Business Suite architecture on AWS](#), World Kinect improved application performance and resiliency. "We saw about a 20 percent performance improvement on different workloads running across Oracle when we moved to AWS, and we saw our mean time to restore [MTTR] go down by 50 percent," says McLean. "Improved infrastructure resiliency is especially critical for our aviation business unit, which is a 24/7 global business."

World Kinect has increased software developer productivity by nearly 400 percent over the past years by increasing software deployments from 100 per quarter to 800 per month. "Through our work with AWS, we've achieved all the objectives we set out with five years ago. We've closed all 22 data centers, we have a scalable platform to support current and future growth, and we can scale our data repository for insights in real time," says Villis.

The company has also built in-house cloud skills by training and certifying employees on AWS services, which is key to the organization's transformation of employee careers. "Collaborating with AWS has helped our employees modernize their skillsets," Villis explains. "Some of them were working in legacy-based architectures, and they're now working on digital approaches and automating solutions. Also, our organization has become a more attractive place to work; people come here because they know they can continually enhance their skillsets while working on cloud-native architectures."

With a modernized application environment on AWS, World Kinect can focus on optimization and innovation. "Ultimately, this migration and training effort is all about digitizing our business and providing capabilities to enter new commercial markets," says Villis. "We're considering doing more for



farms in Europe. Using the weather to predict changes in power demand is one area where we can use advanced analytics on AWS to achieve better business outcomes.”

As World Kinect moves forward with new initiatives, it plans to keep working closely with AWS. “We want to collaborate with people we aspire to be like and look beyond the product. AWS not only offers products and services but also provides timely support, consulting resources, and guidance on our learning journey, including learning plans and various training formats. We look forward to our future together,” concludes McLean.

About World Kinect Corporation

Headquartered in Miami, Florida, World Kinect Corporation (NYSE: WKC) is a global energy distribution and management company, offering a broad suite of solutions across the energy product spectrum. In addition to delivering core energy and fuel solutions, the company is advancing the energy transition by increasing access to renewable energy, lower-carbon fuels, and sustainability advisory services. In more than 200 countries and territories, World Kinect powers the operations of more than 150,000 customers.

AWS Services Used

AWS Enterprise Support

AWS Enterprise Support provides a comprehensive suite of resources, including proactive planning, advisory services, automation tools, communication channels, and 24/7 expert support.

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AWS Professional Services

The AWS Professional Services organization is a global team of experts that can help you realize your desired business outcomes when using the AWS Cloud.

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AWS Training and Certification

Learn from AWS experts. Advance your skills and knowledge. Build your future in the AWS Cloud.



More Energy

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The logo for Jimmy Energy, featuring the word "Jimmy" in a green, rounded, sans-serif font.

Jimmy Energy Builds Scalable and Secure HPC...

Jimmy Energy (Jimmy) has been turning up the heat, with nuclear power taking center stage in its massive decarbonization campaign. Its small-scale generators can reduce carbon footprints across multiple industries, but to facilitate their safe and efficient operation, the company needed a reliable high performance computing (HPC) solution. Using Amazon Web



2024

FRANCE

The logo for Electra, consisting of the word "ELECTRA" in white, uppercase, sans-serif font, centered within a solid red square.

Electra s'appuie sur AWS pour garantir la fiabilité...

Electra, opérateur propriétaire de bornes de recharge ultra-rapide pour véhicules électriques, a choisi AWS dès ses débuts pour assurer la fiabilité, la scalabilité et la sécurité de son infrastructure. En tant que CPO (Charge Point Operator), Electra se distingue par l'installation de bornes ultra-rapides dans les zones urbaines et périurbaines, permettant aux utilisateurs de



2024

UNITED STATES

The logo for Conduit, featuring a stylized green icon of four parallel lines of varying lengths to the left of the word "CONDUIT" in a bold, black, sans-serif font.

Conduit Power Slashed SCADA Deployment Time...

Learn how independent power producer Conduit Power implemented a resilient cloud-based SCADA system using AWS.

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The logo for ENGIE, featuring a blue arc above the word "ENGIE" in a bold, blue, sans-serif font.

ENGIE su AWS

ENGIE è un gruppo energetico globale che lavora per accelerare la transizione verso un mondo a zero emissioni, offrendo energia e servizi a basse emissioni di carbonio. Il gruppo si affida ad Amazon Web Services (AWS) per innovare e raggiungere la sostenibilità per l'azienda e i clienti. Basandosi su AWS, ENGIE promuove la trasformazione digitale con il Common Data Hub, aiuta a proteggere le aziende



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